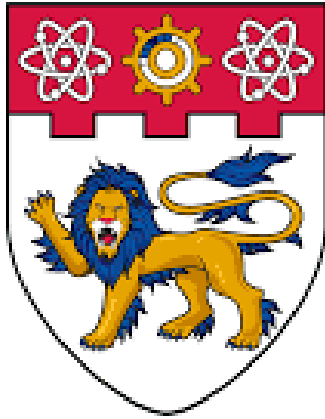




**NANYANG  
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**SINGAPORE**

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**NANYANG TECHNOLOGICAL UNIVERSITY  
(NTU)**

**ReportNow Test Plan (IEEE 829)**

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*Date: 30 October 2025*

*Prepared by: QA Division*

## **Executive Summary**

The ReportNow Test Plan defines the strategy, scope, resources, and schedule required to validate the ReportNow platform's API layer and supporting services. This document complies with IEEE 829 standards and ensures consistent testing practices across all phases of quality assurance.

Testing focuses on backend functionality — from issue reporting and review workflows to administrative operations — to verify that the platform meets its reliability, security, and data-integrity requirements.

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## Introduction

The ReportNow system enables Singapore residents to report municipal issues, track their resolution, and provide feedback. This plan outlines the testing framework that validates the stability, security, and accuracy of the platform's API endpoints, backend logic, and associated libraries.

## Test Items

API Layer: Issues, Reviews, Users, Authentication, Invitations, Admin Activity, Chatbot, Notifications. Libraries include AI Categorisation, Duplicate Detection, and Prisma Schema. Infrastructure includes Jest setup and QA report generation.

## Features to Be Tested

User authentication, profile management, issue lifecycle, reviews, administration, chatbot, notifications, and database schema integrity.

## Features Not to Be Tested

User interface design, performance/load tests beyond Jest, accessibility, localisation, SEO, and external service SLAs.

## Test Approach

Unit tests via Jest, integration tests using Next.js route handlers, schema validation with Prisma mocks, and static review for documentation QA.

## Pass/Fail Criteria

Pass if all tests are green with  $\geq 90\%$  coverage; fail if blocking defects or coverage drops below the threshold.

## Suspension and Resumption Criteria

Testing pauses when essential configs are missing or external APIs change; resumes after fixes and successful suite runs.

## Deliverables

Automated Jest reports, coverage exports, IEEE 829 docs, traceability matrix, and QA logs.

## Testing Tasks

Configure .env.test, initialise Prisma mock DB, execute Jest and lint, review coverage, and log deviations.

## Environmental Needs

16GB RAM laptop, Node v18.18+, mocked Prisma DB, masked credentials in .env.test, Python 3.10+ with python-docx.

## Roles and Responsibilities

PM (Madelyn) approves scope; QA (Yong Huat, Kean Yee) executes; Devs (Wei Ming, Dylan, Seo Hyeon) maintain; Jun Howe manages release.

## Schedule and Milestones

29 Oct 2025: Baseline complete. 30 Oct 2025: Expanded coverage and reports. Post-30 Oct: CI/CD regression integration.

## Entry and Exit Criteria

Entry: merged code and configured env. Exit: full suite pass,  $\geq 90\%$  coverage, updated docs, no Severity 1/2 issues.

## Risk Analysis

External API changes, missing secrets, schema churn, or brittle test data may cause false negatives; mitigated by mocks and checklists.

## Approval

Madelyn (Project Manager), Yong Huat (QA Manager), Wei Ming (Lead Developer).  
Signatures below.